
Priority-based trimmed median filter for removal of high density salt and pepper noise

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Abstract: This paper proposes an efficient and less complex priority-based trimmed median filter algorithm for restoring images corrupted by high density salt and pepper noise. Noisy pixel is replaced by trimmed median value of the horizontal and vertical adjacent four pixels, through this algorithm. If these four are 0's and 255's, then the next priority diagonal adjacent four pixels are used to calculate trimmed median for replacing noisy pixels. If these four are also found as 0's and 255's, then the noisy pixel is left unchanged until the next iteration. Experimental results on different greyscale and colour images show that the proposed algorithm outperforms the standard median filter, adaptive median filter, decision-based algorithm, modified progressive switching median filter and modified decision-based unsymmetric trimmed median filter.

Keywords: high density salt and pepper noise, nonlinear filter, noise detection, median filter, adaptive median filter, unsymmetric trimmed median filter.

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1 Introduction

Impulse noise arises due to the malfunctioning pixels in the camera sensors, transmission in a noisy channel and faulty memory locations in the hardware often pollute the images. Two common types of impulse noise are the salt and pepper noise and random valued noise. The noisy pixels take only the maximum and minimum values in the dynamic range for images corrupted by salt and pepper noise. It is essential to eliminate noise in the images, before starting any other processing such as, edge detection, image segmentation, object recognition and compression. Many approaches have been proposed in the past two decades for this purpose (Acharya and Ray, 2005).

Linear filters blur the edges and smaller details during the denoising process (Kundu et al., 1984; Pitas and Venetsanopoulos, 1986), so, nonlinear filter algorithms are often adopted for the salt-and-pepper noise removal. Among the many nonlinear filter algorithms, standard median filter (MF) based on the statistic ordering is the most typical method for salt and pepper noise that replaces the noisy pixels with median of neighbouring pixels (Astola and Kuosmanen, 1997). Due to its simplicity and efficiency in noise removal, many variants of MFs were proposed (Brownrigg, 1984; Ko and Lee, 1991), but the disadvantage is that it operates on all the pixels in the image and tends to modify good pixels too, thus resulting in loss of finer detail, edge blur and streaking (Chan et al., 2005; Yin et al., 1996). Ideal filtering operation should not modify good pixels so as to avoid blurring and distortion in images (Zhang and Karim, 2002; Eng and Ma, 2001). Adaptive median filter (AMF) (Hwang and Haddad, 1995) replaces only the noise candidates and leaves the good pixels unchanged to avoid damage to good pixels but at high noise densities, the window size increases due to the adaptive nature of the filter. This replaces the noisy pixel with a pixel value that is highly uncorrelated with actual value of the noisy pixel which smears the edges. Progressive switching median filter (PSMF) (Wang and Zhang, 1999) involves two operations namely, impulse detection and noise filtering. Impulse detection phase produces sequence of flag images and greyscale images. Number of flag images and greyscale images depend upon the predefined number of iterations. If the absolute difference between the pixel concerned and the median exceeds a pre-defined threshold value, the corresponding flag bit is set in flag image and greyscale image. These are further modified by the median of a particular neighbourhood, if the flag of current iteration differs from the previous one, else retains the pixel from the previous iterative image. Last iteration flag image is employed for noise filtering the original image but the limitation is time complexity due to the numerous iterations involved both in detection and correction processes and the estimation of optimum threshold also becomes difficult as it may vary from window to window. Thereafter many switching-based median filters that detect noise in first phase and filters the detected noise in second phase were proposed (Chen et al., 1999; Pok et al., 2003; Luo, 2006; Ng and Ma, 2006). Modified progressive switching median (MPSM) filter (Kuykin et al., 2009), which is also based on switching scheme detects the noisy pixel in the impulse detection phase by inspecting whether the processing pixel is

equal to the maximum and minimum pixel values within the window. If so, the pixel is considered to be noisy, else noise free because noise detection is affected by the size of variable window.

Decision-based algorithm (DBA) was proposed to overcome the above drawback (Srinivasan and Ebenezer, 2007) and denoising process uses a 3×3 window in this algorithm. If the processing pixel value is 0 or 255 minimum or maximum respectively in the dynamic range, it is processed or else it is left unchanged. The median value will be 0 or 255 at high noise density, which is noisy. In order to avoid this drawback, decision-based unsymmetric trimmed median filter (DBUTMF) was proposed (Aiswarya et al., 2010). The trimmed median value cannot be obtained, if the selected window contains all 0's or 255's or both at high noise densities and so, this algorithm does not give better results at high noise density. Modified decision-based unsymmetric trimmed median filter (MDBUTMF) (Esakkirajan et al., 2011) algorithm works well in this case. If the selected window contains all 0's or 255's or both, then the processed pixel is replaced with mean of selected window which removes the drawback of DBUTMF, but the mean value of all 0's or 255's or both will be an error value.

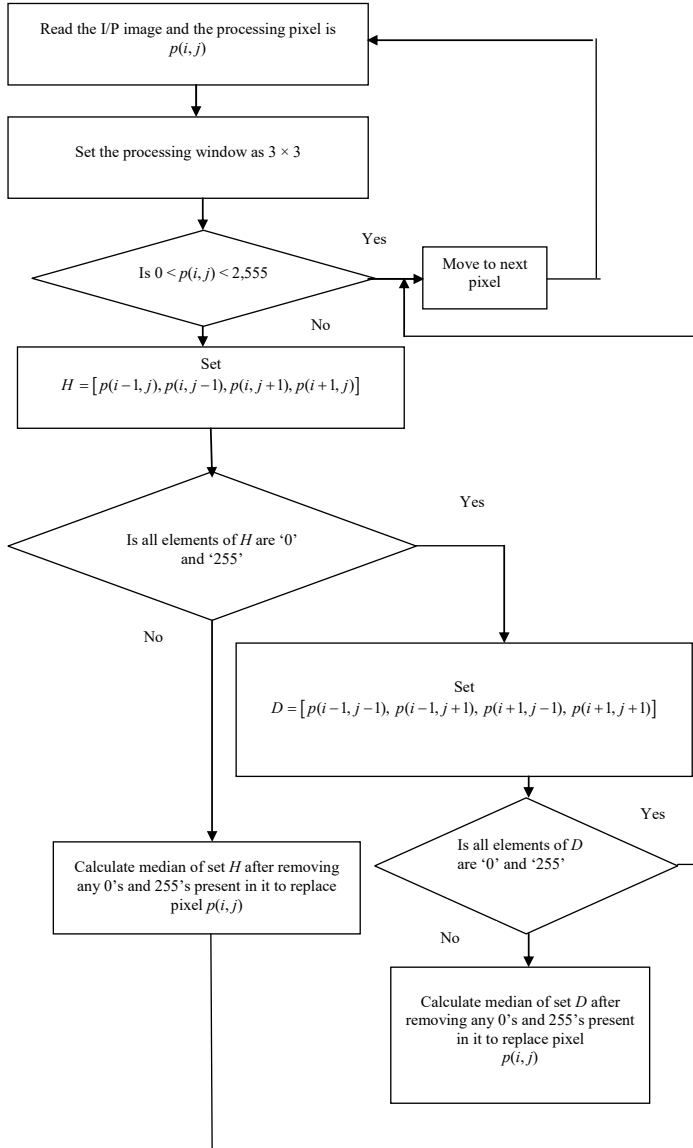
These various median-based adaptive filters has the following limitations such as, various degrees of complexity, design approaches varies between noise suppression and detail preservation and the quality of the results are of various degrees due to the difficulties in choosing the most appropriate median. The proposed filtering algorithm is simple as it uses only four elements instead of eight. Although few other algorithms use more elements, the proposed algorithm dramatically outperforms all other existing algorithms.

Rest of the portions of this paper are organised in the following order. The priority-based trimmed median filter is formulated in Section 2, Section 3 and Section 4 deals with results and discussions. Finally conclusions are drawn in Section 5.

2 Priority-based trimmed median filter algorithm

If the processing pixel is between maximum and minimum grey level value in the dynamic range, it is left unchanged in this proposed priority-based trimmed median filter (PBTMF) algorithm but if the processing pixel is in maximum or minimum grey level value in the dynamic range, it is processed by taking a 3×3 mask. Among the eight elements of the processing pixel within the mask, four horizontal and vertical pixels that are adjacent are taken for calculating the trimmed median which replaces the noisy pixel. If all these four elements are noisy, then diagonally adjacent second priority four pixels are taken for computing trimmed median. In worst case, if these four pixels are also 0's and 255's, the noisy pixel is left unchanged for next iteration. The window size is fixed to 3×3 only because every pixel has high correlation with the adjacent pixels only and not on any other pixel in the window of higher dimension. The steps of PBTMF are as follows:

Figure 1 Flow chart



2.1 Algorithm

- Step 1 The pixel being processed is denoted by $p(i, j)$ and a 3×3 mask is used for processing.
- Step 2 If $p(i, j)$ is a value between 0 and 255, then the pixel value is left unchanged and goes to step 7.
- Step 3 If $p(i, j)$ is 0 or 255 then, select a set ' H ' such that $H = [p(i-1, j), p(i, j-1), p(i, j+1), p(i+1, j)]$. If all the elements of H are 0's and 255's then move to step 5 else to step 4.
- Step 4 Calculate median of set H after removing any 0's and 255's present in it to replace pixel $p(i, j)$ and move to step 7.
- Step 5 Select a set ' D ' such that $D = [p(i-1, j-1), p(i-1, j+1), p(i+1, j-1), p(i+1, j+1)]$. If all the elements of D are 0's and 255's then $p(i, j)$ is unchanged and go to step 7, else go to step 6.
- Step 6 Calculate median of set ' D ' after removing any 0's and 255's present in it to replace pixel $p(i, j)$.
- Step 7 Move to next pixel and repeat steps 1 to 6, until all the pixels in the image are processed.
- Step 8 A second round of iteration is done if few noisy pixels are still present in image.

Eight adjacent pixels were split in to two sets, H and D in the above algorithm and first priority was given to H set during the process because pixels in H set were found to have more correlation with the centre pixel than the D set. Moreover, by splitting it in to two sets, only four pixels are processed at a time, which reduces processing time and makes physical realisation simple. Different cases faced by the algorithm are illustrated through examples in the next section.

2.2 Illustration

In the above given algorithm, there are many IF-THEN statements. Every IF statement is an encountered case during denoising. These different cases are elucidated below through examples.

- Case 1 Pixel $p(i, j)$, which is processed is a value between 0 and 255:

$$\begin{bmatrix} 96 & 95 & 97 \\ 0 & 97 & 95 \\ 94 & 255 & 0 \end{bmatrix}$$

In this case $p(i, j)$, which is 97 is left unchanged as it falls between 0 and 255.

- Case 2 The pixel to be processed is either 0 or 255 and set H contains noise.

$$\begin{bmatrix} 98 & 96 & 93 \\ 255 & 255 & 92 \\ 255 & 94 & 0 \end{bmatrix}$$

In this case $p(i, j)$ is noisy, so from the set H , eliminate any 0 or 255 present. After elimination the set becomes [92, 94, 96], whose median is 94 and this value replaces the centre pixel.

Case 3 The pixel to be processed is either 0 or 255 and set H contains only noise.

$$\begin{bmatrix} 98 & 0 & 96 \\ 255 & 255 & 0 \\ 97 & 0 & 255 \end{bmatrix}$$

Since set H is all noisy, select set D and remove any 0 or 255 from it. Now, set D becomes [96, 97, 98], whose median is 97 and this value replaces the processing pixel $p(i, j)$.

Case 4 Pixel $p(i, j)$ is noisy and set H as well as D contains only 0's and 255's.

$$\begin{bmatrix} 0 & 0 & 255 \\ 255 & 255 & 0 \\ 0 & 0 & 255 \end{bmatrix}$$

$p(i, j)$ is left unchanged and is mended in the second iteration. The experimental results showed that this case was the rarest for even 90% of noise density which explains that all the pixels were repaired in the first iteration itself. The entire algorithm is given in the flow chart shown in Figure 1.

3 Experimental results

Fifteen standard greyscale and colour images such as Lena, Baboon, Peppers, cameraman, etc., were subjected to the proposed algorithm. The results of the proposed algorithm was compared with many other algorithms such as MF, AMF, DBA, MPSM and most recently proposed MDBUTMF for validation.

Quantitative analysis was done through the metrics peak signal to noise ratio (PSNR) and structural similarity index (SSIM) (Wang et al., 2004) as given in the equation (1) and (3).

$$PSNR \text{ in } dB = 10 \log_{10} \left(\frac{255^2}{MSE} \right) \quad (1)$$

$$MSE = \frac{\sum_i \sum_j (X(i, j) - Y(i, j))^2}{M \times N} \quad (2)$$

where MSE stands for mean squared error, X is the original image, Y is the restored image and $M * N$ is the size of the image.

$$SSIM(X, Y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3)$$

$$\mu_x = \frac{1}{M \times N} \sum_{i=1}^{MN} x_i \quad (4)$$

$$\sigma_x = \sqrt{\left(\frac{1}{M \times N} \sum_{i=1}^{MN} (x_i - \mu_x)^2 \right)} \quad (5)$$

$$\sigma_{xy} = \frac{1}{M \times N} \sum_{i=1}^{MN} (x_i - \mu_x)(y_i - \mu_y) \quad (6)$$

In the above said equations (4) and (5), μ is the mean, σ is the standard deviation, $C1 = (K_1L)^2$ and $C2 = (K_2L)^2$. L is the dynamic range of pixel values (255 for 8-bit greyscale image) and $K_1, K_2 \ll 1$. (In our experiment, we used $K_1 = 0.01$ and $K_2 = 0.03$)

4 Discussions

Run time analysis was done for all the algorithms using MATLAB 7.10 software on a PC with 2.8 GHz CPU and 1 GB RAM. Tables 1 and 2 show the comparison of PSNR and SSIM values of the proposed algorithm against the existing algorithms for standard Lena image at different noise densities ranging from 10% to 90%. Tables 3 and 4 show the results for Baboon. Table 5 gives the computation time of all the algorithms. It was noted that the proposed algorithm performed consistently over the entire range of noise density. With respect to recently proposed MDBUTMF (Esakkirajan et al., 2011) algorithm, we have on an average 7% improvement in PSNR spread over entire range of noise density and 43% reduction in computation time. Computation time is marginally greater than MF, but it is much affordable for a qualitative performance.

Table 1 Comparison of PSNR values of various algorithms for standard ‘Lena’ image at different noise densities

Noise		PSNR in dB				
Density (%)	MF	AMF	DBA	MPSPM	MDBUTMF	Proposed
10	29.22	30.39	35.12	36.56	37.49	39.17
20	25.42	26.01	32.61	34.12	34.80	36.50
30	24.30	25.03	28.61	32.35	33.57	34.81
40	20.88	21.72	24.33	30.36	31.10	32.21
50	17.60	18.89	20.72	27.11	29.40	31.01
60	14.92	15.90	17.57	25.12	28.12	29.75
70	12.73	13.47	15.07	21.85	26.40	28.24
80	10.82	11.49	13.54	18.75	23.90	26.50
90	8.77	9.46	11.48	17.31	20.65	22.85

Table 2 Comparison of SSIM index of various algorithms for standard ‘Lena’ image at different noise densities

<i>Noise</i>		<i>SSIM index</i>				
<i>Density (%)</i>	<i>MF</i>	<i>AMF</i>	<i>DBA</i>	<i>MPSM</i>	<i>MDBUTMF</i>	<i>Proposed</i>
10	.9157	.9232	.9517	.9638	.9855	.9960
20	.9054	.9138	.9432	.9588	.9806	.9924
30	.8574	.8941	.9081	.9414	.9733	.9873
40	.7209	.8373	.8960	.9326	.9673	.9820
50	.6462	.6851	.7869	.9082	.9572	.9754
60	.4882	.5575	.6241	.8975	.9428	.9670
70	.3081	.3800	.4370	.8785	.9291	.9551
80	.1649	.2130	.2591	.7527	.8832	.9296
90	.0727	.0855	.1107	.5951	.7443	.8691

Table 3 Comparison of PSNR values of various algorithms for standard ‘Baboon’ image at different noise densities

<i>Noise</i>		<i>PSNR in dB</i>				
<i>Density (%)</i>	<i>MF</i>	<i>AMF</i>	<i>DBA</i>	<i>MPSM</i>	<i>MDBUTMF</i>	<i>Proposed</i>
10	28.49	29.22	34.22	36.48	36.98	39.90
20	25.76	26.95	33.55	34.71	35.56	37.46
30	23.06	25.31	28.12	31.02	33.63	35.27
40	20.72	22.57	25.60	30.44	31.94	33.04
50	17.54	18.91	21.18	28.87	29.74	31.53
60	13.19	14.40	18.69	25.63	28.60	30.38
70	11.03	13.58	16.62	22.68	26.19	28.13
80	10.18	11.77	14.71	19.52	24.06	27.51
90	9.57	10.31	12.35	17.69	20.26	23.60

Table 4 Comparison of SSIM index of various algorithms for standard ‘Baboon’ image at different noise densities

<i>Noise</i>		<i>SSIM index</i>				
<i>Density (%)</i>	<i>MF</i>	<i>AMF</i>	<i>DBA</i>	<i>MPSM</i>	<i>MDBUTMF</i>	<i>Proposed</i>
10	.9096	.9224	.9509	.9629	.9848	.9981
20	.9015	.9195	.9465	.9591	.9812	.9937
30	.8297	.8965	.9017	.9495	.9764	.9881
40	.7195	.8403	.8975	.9303	.9701	.9834
50	.6348	.6884	.7903	.9143	.9675	.9768
60	.4680	.5493	.6427	.9038	.9527	.9722
70	.2914	.3567	.4804	.8876	.9167	.9475
80	.1642	.2175	.3149	.7879	.8965	.9325
90	.0881	.0974	.1382	.6018	.7215	.8850

Table 5 Comparison of run time of various algorithms for standard 'Lena' image at different noise densities

Noise		Run time (sec)				
Density (%)	MF	AMF	DBA	MPSM	MDBUTMF	Proposed
10	1.998	10.709	13.283	23.733	4.718	3.088
20	2.395	11.001	14.782	25.118	8.253	4.834
30	2.874	11.171	17.250	27.545	11.854	6.212
40	3.243	11.327	18.203	28.096	13.083	7.182
50	3.837	11.449	18.737	30.475	15.766	8.891
60	3.983	11.451	20.143	31.086	15.933	9.592
70	4.281	11.690	20.967	34.658	18.075	9.878
80	4.683	11.837	21.063	35.985	19.814	10.612
90	5.036	11.967	22.521	37.396	19.924	11.592

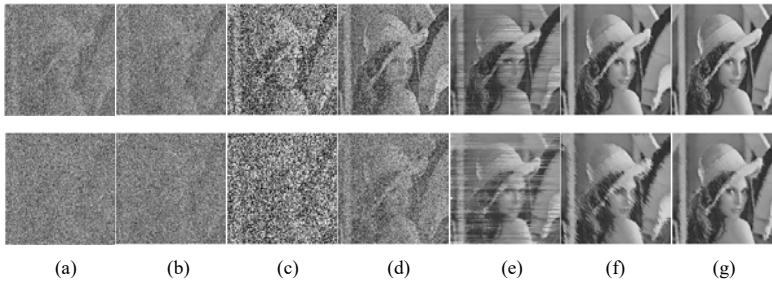
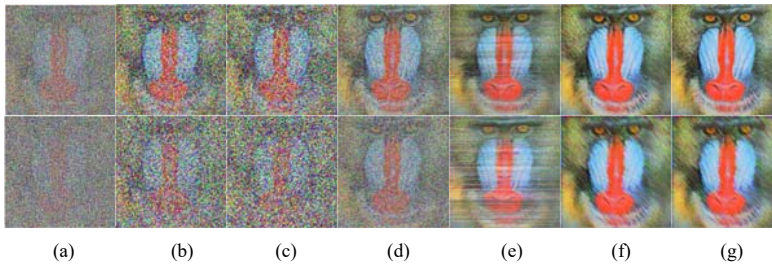
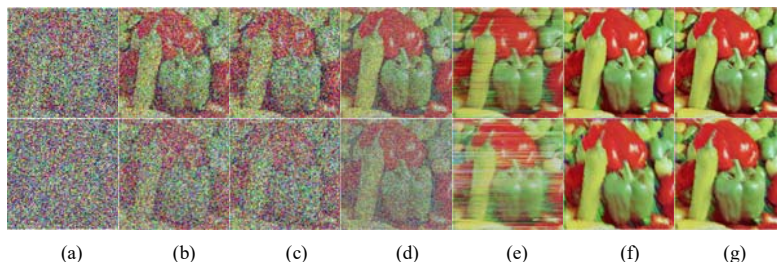
Figure 2 Restoration results of various median filters for Lena image. Row 1 and row 2 show the filtered images of various denoising algorithms for image corrupted by 80% and 90% noise density respectively, (a) corrupted image (b) output of MF (c) output of AMF (d) output of DBA (e) output of MPSM (f) output of MDBUTMF (g) output of PBTMF**Figure 3** Restoration results of various median filters for Baboon image. Row 1 and Row 2 show the filtered images of various denoising algorithms for image corrupted by 80% and 90% noise density respectively, (a) corrupted image (b) output of MF (c) output of AMF (d) output of DBA (e) output of MPSM (f) output of MDBUTMF (g) output of PBTMF (see online version for colours)

Figure 4 Restoration results of various median filters for Peppers image. Row 1 and Row 2 show the filtered images of various denoising algorithms for image corrupted by 80% and 90% noise density respectively, (a) corrupted image (b) output of MF (c) output of AMF (d) output of DBA (e) output of MPSM (f) output of MDBUTMF (g) output of PBTMF (see online version for colours)



Qualitative analysis of the proposed algorithm versus various other algorithms for standard Lena, Baboon and Peppers image is shown in Figure 2, Figure 3 and Figure 4 respectively. In Figures 2, 3, 4, first column contains noisy image and subsequent columns show the results of MF, AMF, DBA, MPSM, MDBUTMF and PBTMF. It can be perceived that quality of image filtered through the proposed algorithm is better than the images filtered by the existing algorithms. Normally in any image processing paper, if the preprocessing (Bharathi et al., 2014) stage demands for salt and pepper noise removal our algorithm is a best substitute for the same.

5 Conclusions

An efficient and simple denoising algorithm for salt and pepper noise is proposed in this paper. The algorithm uses only the adjacent pixels of corrupted pixel that have higher correlation, which provides good edge preservation. Since four inputs are taken at a time for denoising a pixel, the algorithm requires less computation time and simple hardware realisation structure which makes it an apt option for introducing into devices that work under noisy environment. Efficiency of the algorithm was substantiated through quantitative and qualitative analysis. This algorithm consistently performed better than the existing algorithms for noise densities ranging from 10% to 90%.

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